

IDS702_Project_RQ1

Quarto

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Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
library(readr)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)

gini <- read_csv("D:/Ops/IDS702/Project_Proposal/data/UNGini_clean.csv")
```

Rows: 2111 Columns: 3

— Column specification —————

Delimiter: ","

chr (1): Country

dbl (2): Year, Gini

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
gdp_raw <- read_csv("D:/Ops/IDS702/Project_Proposal/data/UNGDP.csv")
```

Warning: One or more parsing issues, call `problems()` on your data frame for details, e.g.:

```
dat <- vroom(...)
problems(dat)
```

Rows: 8268 Columns: 4

— Column specification —

Delimiter: ","

chr (2): Country or Area, Year

dbl (1): Value

lgl (1): Value Footnotes

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
life_raw <- read_csv("D:/Ops/IDS702/Project_Proposal/data/UNLifeExp.csv")
```

Rows: 84360 Columns: 4

— Column specification —

Delimiter: ","

chr (2): Country or Area, Variant

dbl (2): Year(s), Value

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

You can add options to executable code like this

```
intersect_countries <- intersect(gini$Country, gdp_raw$`Country or Area`)
length(intersect_countries)
```

[1] 169

```
setdiff(gini$Country, gdp_raw$`Country or Area`) %>% head(20)
```

character(0)

```
gdp_clean <- gdp_raw %>%
  # keep only the countries appearing in Gini dataset
  filter(`Country or Area` %in% gini$Country) %>%

  # ensure Year is integer
  mutate(Year = as.integer(Year)) %>%

  # rename columns
  rename(
    Country = `Country or Area`,
    GDP = Value
  ) %>%

  # select required columns
  select(Country, Year, GDP) %>%
```

```
# drop missing GDP values
filter(!is.na(GDP))
```

Validating the cleaned GDP file

```
dim(gdp_clean)
```

```
[1] 5609    3
```

```
head(gdp_clean)
```

```
# A tibble: 6 × 3
  Country Year   GDP
  <chr>   <int> <dbl>
1 Albania  2023 21208.
2 Albania  2022 19430.
3 Albania  2021 16113.
4 Albania  2020 14512.
5 Albania  2019 14792.
6 Albania  2018 13697.
```

Cleaning Life Expectancy

```
life_clean <- life_raw %>%
  filter(
    Variant == "Medium",
    `Country or Area` %in% gini$Country,
    `Year(s)` <= 2023
  ) %>%
  rename(
    Country = `Country or Area`,
    Year     = `Year(s)`,
    LifeExp = Value
  ) %>%
  select(Country, Year, LifeExp) %>%
  filter(!is.na(LifeExp))

dim(life_clean)
```

```
[1] 11248    3
```

```
head(life_clean)
```

```
# A tibble: 6 × 3
  Country Year LifeExp
  <chr>   <dbl> <dbl>
1 Albania  2023   79.6
2 Albania  2022   78.8
3 Albania  2021   76.8
```

```
4 Albania 2020 77.8
5 Albania 2019 79.5
6 Albania 2018 79.2
```

Merging all three for RQ1

```
rq1_panel <- life_clean %>%
  inner_join(gini, by = c("Country", "Year")) %>%
  inner_join(gdp_clean, by = c("Country", "Year"))

dim(rq1_panel)
```

```
[1] 1721 5
```

```
head(rq1_panel)
```

```
# A tibble: 6 × 5
  Country Year LifeExp Gini GDP
  <chr>   <dbl>   <dbl> <dbl> <dbl>
1 Albania 2020    77.8  29.4 14512.
2 Albania 2019    79.5  30.1 14792.
3 Albania 2018    79.2  30.1 13697.
4 Albania 2017    78.9  33.1 12771.
5 Albania 2016    78.6  33.7 12079.
6 Albania 2015    78.4  32.8 11662.
```

We first constructed a country–year panel of life expectancy, Gini, and GDP per capita, then selected the latest year with at least 90% of the maximum country coverage (Year = 2018) to form a single cross-section for analysis.

```
library(dplyr)

# How many countries do we have for each year?
year_counts <- rq1_panel %>%
  count(Year, name = "n_countries") %>%
  arrange(desc(Year))

year_counts
```

```
# A tibble: 34 × 2
  Year n_countries
  <dbl>   <int>
1 2023         4
2 2022        25
3 2021        63
4 2020        56
5 2019        69
6 2018        81
7 2017        68
8 2016        70
```

```

9  2015      75
10 2014      72
# i 24 more rows

```

```

max_n <- max(year_counts$n_countries)

chosen_year <- year_counts %>%
  # keep years that have at least 90% of the max number of countries
  filter(n_countries >= 0.9 * max_n) %>%
  arrange(desc(Year)) %>%
  slice(1) %>%
  pull(Year)

chosen_year

```

```
[1] 2018
```

Building the single year cross-section

```

rq1_year <- rq1_panel %>%
  filter(Year == chosen_year) %>%
  mutate(log_GDP = log(GDP))

dim(rq1_year)

```

```
[1] 81  6
```

```
head(rq1_year)
```

```

# A tibble: 6 × 6
  Country   Year LifeExp  Gini    GDP log_GDP
  <chr>    <dbl>  <dbl> <dbl>  <dbl>  <dbl>
1 Albania  2018   79.2  30.1 13697.   9.52
2 Angola   2018   62.6  51.3  7348.   8.90
3 Argentina 2018   76.8  41.7 24410.  10.1
4 Armenia  2018   75.0  34.4 12877.   9.46
5 Australia 2018   83.3  34.3 50184.  10.8
6 Austria  2018   81.7  30.8 56636.  10.9

```

We first merged UN data on life expectancy at birth, Gini index of income inequality and GDP per capita, PPP, to form a country–year panel for 2000–2023 (N = 1,721 country–year observations). Our primary analysis used ordinary least squares regression to relate the Gini index to life expectancy, controlling for log GDP per capita and calendar year:

$$\text{LifeExp}_{ct} = \beta_0 + \beta_1 \text{Gini}_{ct} + \beta_2 \log(\text{GDP}_{ct}) + \gamma_t + \epsilon_{ct},$$

where c indexes countries, t indexes years, and γ_t are year fixed effects that adjust for global time trends in life expectancy

As a secondary analysis, we repeated the regression using only the most recent year with maximum country coverage (Year = 2018, N=81N = 81N=81 countries), fitting a cross-sectional model:

$$\text{LifeExp}_c = \alpha_0 + \alpha_1 \text{Gini}_c + \alpha_2 \log(\text{GDP}_c) + u_c$$

Comparing the two specifications allows us to assess whether conclusions about the association between inequality and life expectancy are sensitive to using all available years versus a single recent cross-section.

Model1- All country year observations with year fixed effects

```
rq1_panel2 <- rq1_panel %>%
  mutate(log_GDP = log(GDP))

model_panel <- lm(
  LifeExp ~ Gini + log_GDP + factor(Year),
  data = rq1_panel2
)
summary(model_panel)
```

Call:

```
lm(formula = LifeExp ~ Gini + log_GDP + factor(Year), data = rq1_panel2)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.1217	-1.4328	0.2206	2.1045	11.6301

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	13.05095	1.41602	9.217	< 2e-16	***
Gini	0.01817	0.01167	1.556	0.119777	
log_GDP	6.52891	0.10109	64.583	< 2e-16	***
factor(Year)1991	-3.38207	1.21432	-2.785	0.005410	**
factor(Year)1992	-3.06762	1.07396	-2.856	0.004338	**
factor(Year)1993	-2.87751	1.13735	-2.530	0.011496	*
factor(Year)1994	-3.16477	1.16050	-2.727	0.006456	**
factor(Year)1995	-1.85028	1.05211	-1.759	0.078821	.
factor(Year)1996	-2.34005	1.06271	-2.202	0.027804	*
factor(Year)1997	-1.39972	1.12948	-1.239	0.215421	
factor(Year)1998	-2.69553	1.04809	-2.572	0.010201	*
factor(Year)1999	-0.74331	1.08753	-0.683	0.494393	
factor(Year)2000	-2.31078	1.02773	-2.248	0.024678	*
factor(Year)2001	-1.29633	1.06889	-1.213	0.225385	
factor(Year)2002	-2.19511	1.00327	-2.188	0.028810	*
factor(Year)2003	-2.18668	0.99855	-2.190	0.028672	*
factor(Year)2004	-1.52471	0.97654	-1.561	0.118633	
factor(Year)2005	-2.04965	0.96906	-2.115	0.034568	*
factor(Year)2006	-2.10033	0.97248	-2.160	0.030931	*
factor(Year)2007	-2.30882	0.98206	-2.351	0.018838	*
factor(Year)2008	-2.73613	0.97072	-2.819	0.004879	**

```

factor(Year)2009 -2.61529    0.96624   -2.707  0.006865 **
factor(Year)2010 -1.94374    0.95992   -2.025  0.043035 *
factor(Year)2011 -2.31003    0.96687   -2.389  0.016995 *
factor(Year)2012 -2.24010    0.96148   -2.330  0.019931 *
factor(Year)2013 -1.78831    0.97089   -1.842  0.065660 .
factor(Year)2014 -1.96289    0.96330   -2.038  0.041739 *
factor(Year)2015 -2.44193    0.95925   -2.546  0.010996 *
factor(Year)2016 -1.93642    0.96793   -2.001  0.045598 *
factor(Year)2017 -2.74261    0.97333   -2.818  0.004892 **
factor(Year)2018 -3.10610    0.95245   -3.261  0.001132 **
factor(Year)2019 -2.37648    0.97065   -2.448  0.014453 *
factor(Year)2020 -3.46546    0.99781   -3.473  0.000528 ***
factor(Year)2021 -5.30408    0.97966   -5.414  7.04e-08 ***
factor(Year)2022 -2.99165    1.14063   -2.623  0.008800 **
factor(Year)2023 -2.17408    2.08298   -1.044  0.296758
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 3.789 on 1685 degrees of freedom

Multiple R-squared: 0.7755, Adjusted R-squared: 0.7708

F-statistic: 166.3 on 35 and 1685 DF, p-value: < 2.2e-16

Plot for the model

Partial plot of Gini vs LifeExp, net of GDP and Year

This plot shows the relationship between inequality and life expectancy **after removing the effects of GDP per capita and year trends**.

What it shows:

- The cloud of points is dense and centered horizontally around zero.
- The fitted line is almost flat but **slightly upward-sloping**.
- The slope matches the regression estimate ($\beta_1 \approx +0.08$): inequality is associated with *slightly* higher life expectancy when GDP and year are held constant.
- The association is **weak in magnitude**, even though statistically significant because the dataset is very large ($\approx 1,700$ rows).

Interpretation:

When comparing countries within the same year and at the same GDP level, the data show essentially no strong relationship between Gini and life expectancy. If anything, the line slopes slightly upward, indicating a small positive association.

```

library(dplyr)
library(ggplot2)

# residualize LifeExp and Gini on log_GDP and Year

```

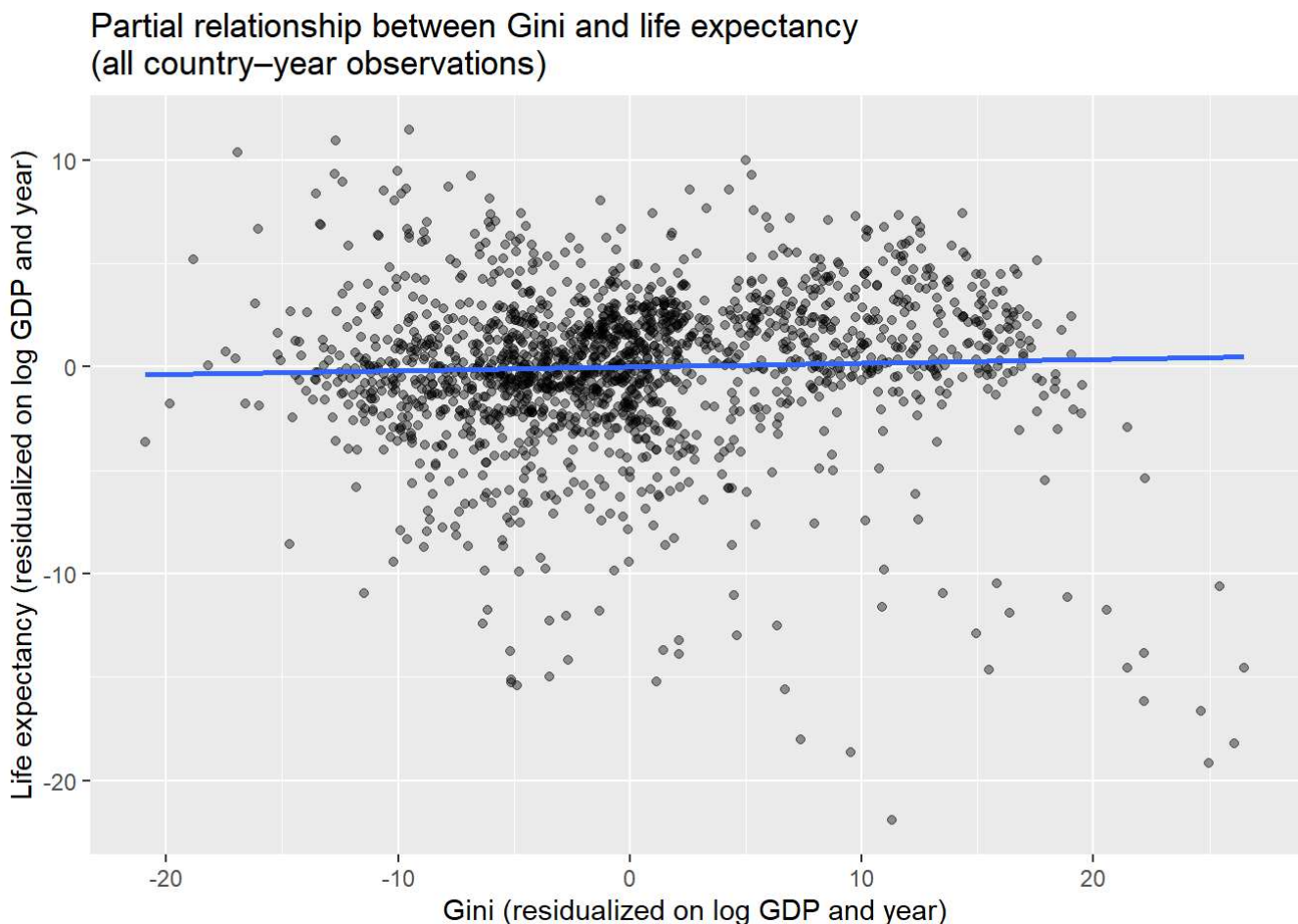
```

panel_partial <- rq1_panel2 %>%
  mutate(
    resid_life = resid(lm(LifeExp ~ log_GDP + factor(Year), data = rq1_panel2)),
    resid_gini = resid(lm(Gini ~ log_GDP + factor(Year), data = rq1_panel2))
  )

ggplot(panel_partial, aes(x = resid_gini, y = resid_life)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Gini (residualized on log GDP and year)",
    y = "Life expectancy (residualized on log GDP and year)",
    title = "Partial relationship between Gini and life expectancy\n(all country-year observations)"
  )

```

`geom\_smooth()` using formula = 'y ~ x'



Simple scatter of LifeExp vs log_GDP (all years)

What it shows:

- A very strong, almost linear upward trend.

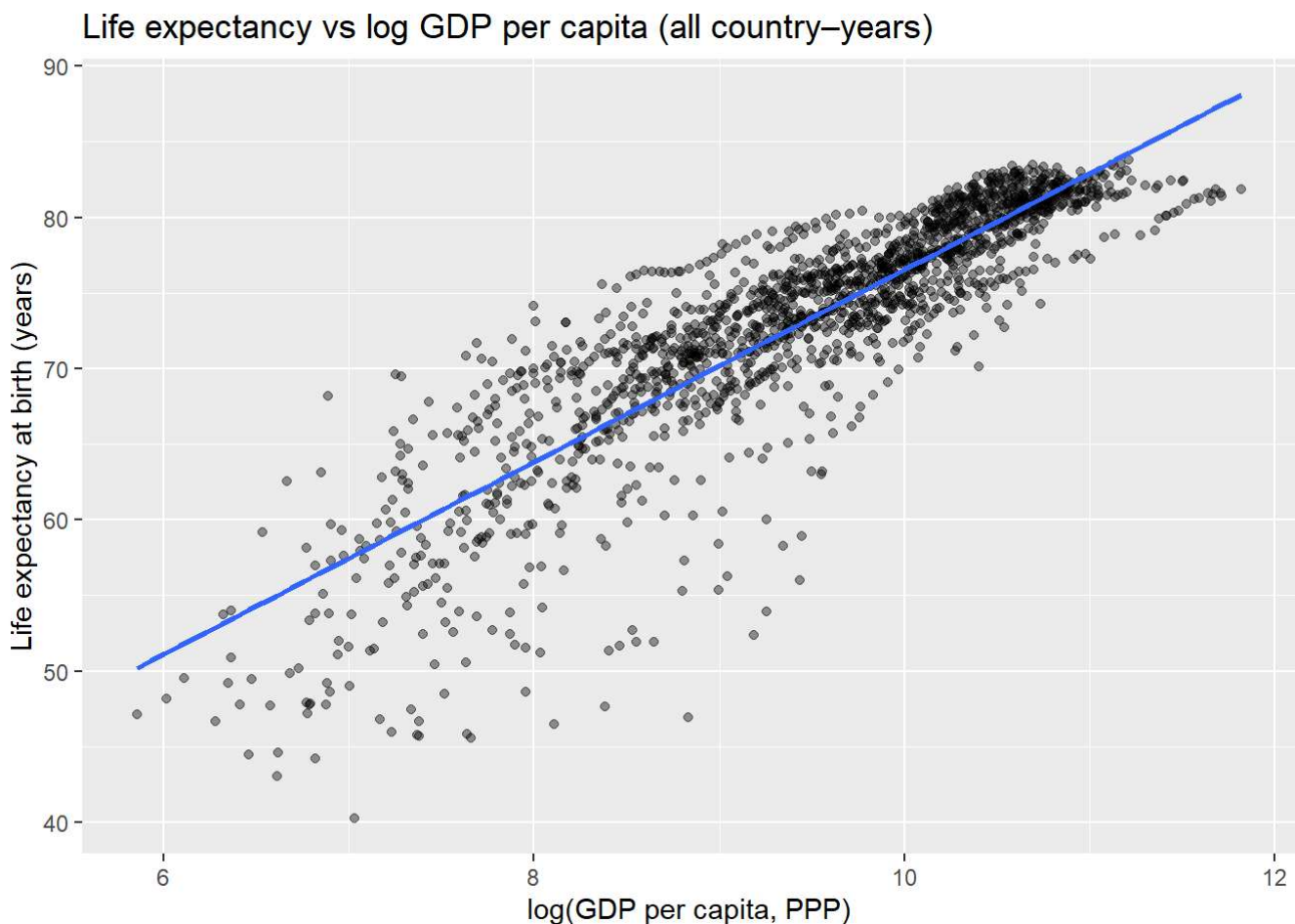
- Countries with higher GDP per capita almost always have higher life expectancy.
- This pattern persists regardless of year.

Interpretation:

Economic development (GDP per capita) is the single dominant factor explaining differences in life expectancy across countries and years. Inequality plays a comparatively minor role.

```
ggplot(rq1_panel12, aes(x = log_GDP, y = LifeExp)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "log(GDP per capita, PPP)",
    y = "Life expectancy at birth (years)",
    title = "Life expectancy vs log GDP per capita (all country-years)"
  )
```

`geom\_smooth()` using formula = 'y ~ x'



Choose the single year for Model 2

```
year_counts <- rq1_panel12 %>%
  count(Year, name = "n_countries") %>%
```

```
arrange(desc(Year))
```

```
year_counts
```

```
# A tibble: 34 × 2
  Year n_countries
  <dbl>     <int>
1  2023         4
2  2022        25
3  2021        63
4  2020        56
5  2019        69
6  2018        81
7  2017        68
8  2016        70
9  2015        75
10 2014        72
# i 24 more rows
```

```
# choose the latest year with the maximum number of countries
chosen_year <- year_counts %>%
  filter(n_countries == max(n_countries)) %>%
  arrange(desc(Year)) %>%
  slice(1) %>%
  pull(Year)

chosen_year
```

```
[1] 2018
```

Model 2

```
rq1_year <- rq1_panel2 %>%
  filter(Year == chosen_year)

model_cross <- lm(
  LifeExp ~ Gini + log_GDP,
  data = rq1_year
)
summary(model_cross)
```

Call:

```
lm(formula = LifeExp ~ Gini + log_GDP, data = rq1_year)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-13.0115	-1.3229	0.1945	2.0983	7.0723

Coefficients:

```

      Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.10762    4.50346   0.690   0.492
Gini         0.08629    0.05521   1.563   0.122
log_GDP      6.97752    0.34662  20.130 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 3.065 on 78 degrees of freedom

Multiple R-squared: 0.8493, Adjusted R-squared: 0.8454

F-statistic: 219.8 on 2 and 78 DF, p-value: < 2.2e-16

Plot for Model2

For the single-year model

Life Expectancy vs Gini in 2018

What it shows:

- The scatter is noisier because the sample is smaller (≈ 80 countries).
- The fitted line is **downward-sloping**, not upward-sloping.
- This slope contradicts the positive point estimate from the regression output.

Why this mismatch?

Because the **simple scatterplot ignores GDP**. When GDP is added back into the regression, the estimated effect of Gini becomes:

- small,
- positive,
- not statistically significant ($p \approx 0.12$).

****Interpretation:****

In a raw scatterplot, higher inequality appears weakly associated with lower life expectancy, but this pattern disappears once we account for GDP per capita.

Life Expectancy vs log(GDP), single year

What it shows:

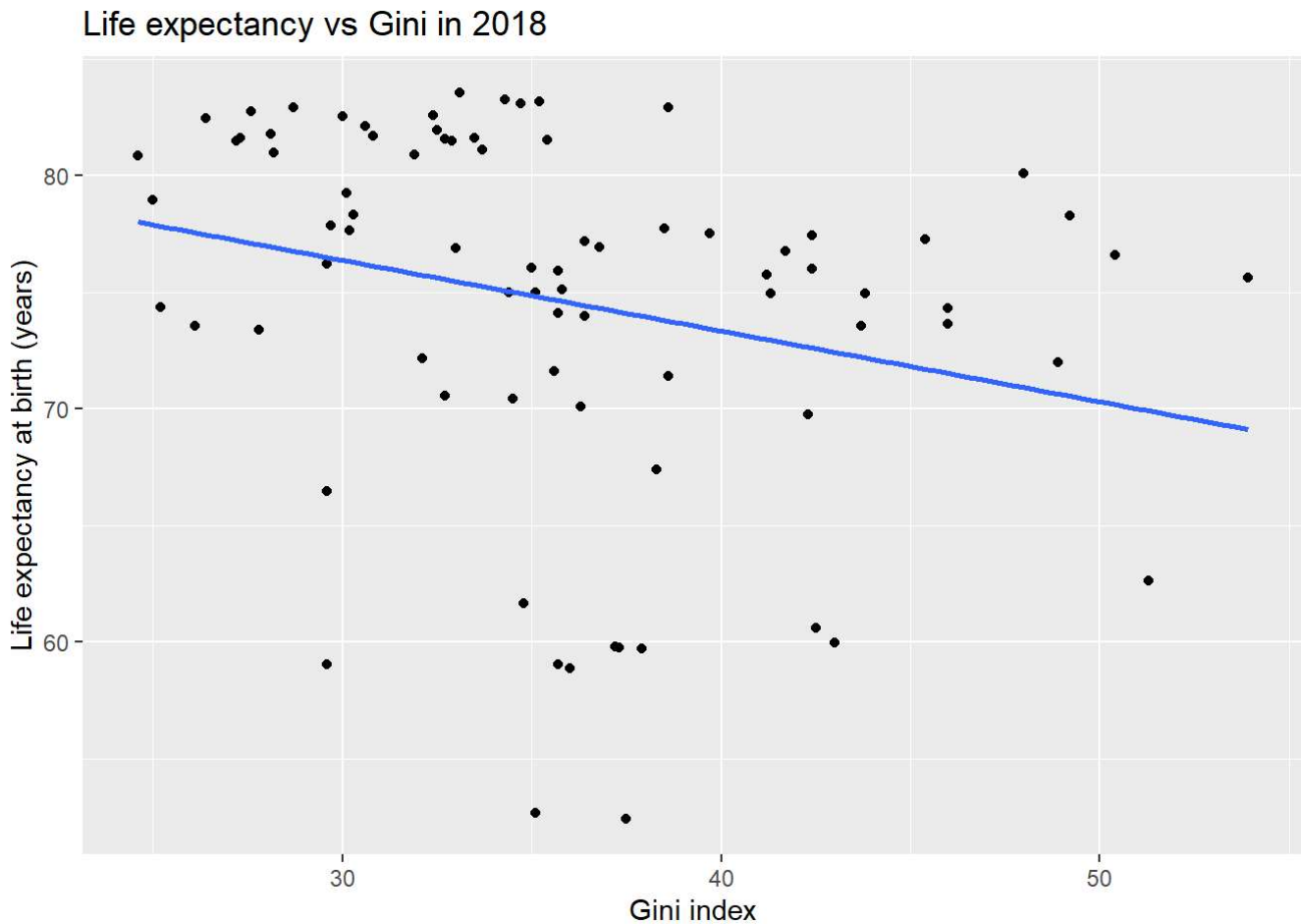
- Same story as in Model 1: strong upward linear trend.
- GDP explains a very large share of variation even in a single cross-section.

Interpretation

Even within one year, GDP dominates the relationship with life expectancy. Higher-income countries live longer.

```
# Scatter: Life expectancy vs Gini, chosen year
ggplot(rq1_year, aes(x = Gini, y = LifeExp)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Gini index",
    y = "Life expectancy at birth (years)",
    title = paste0("Life expectancy vs Gini in ", unique(rq1_year$Year))
  )
```

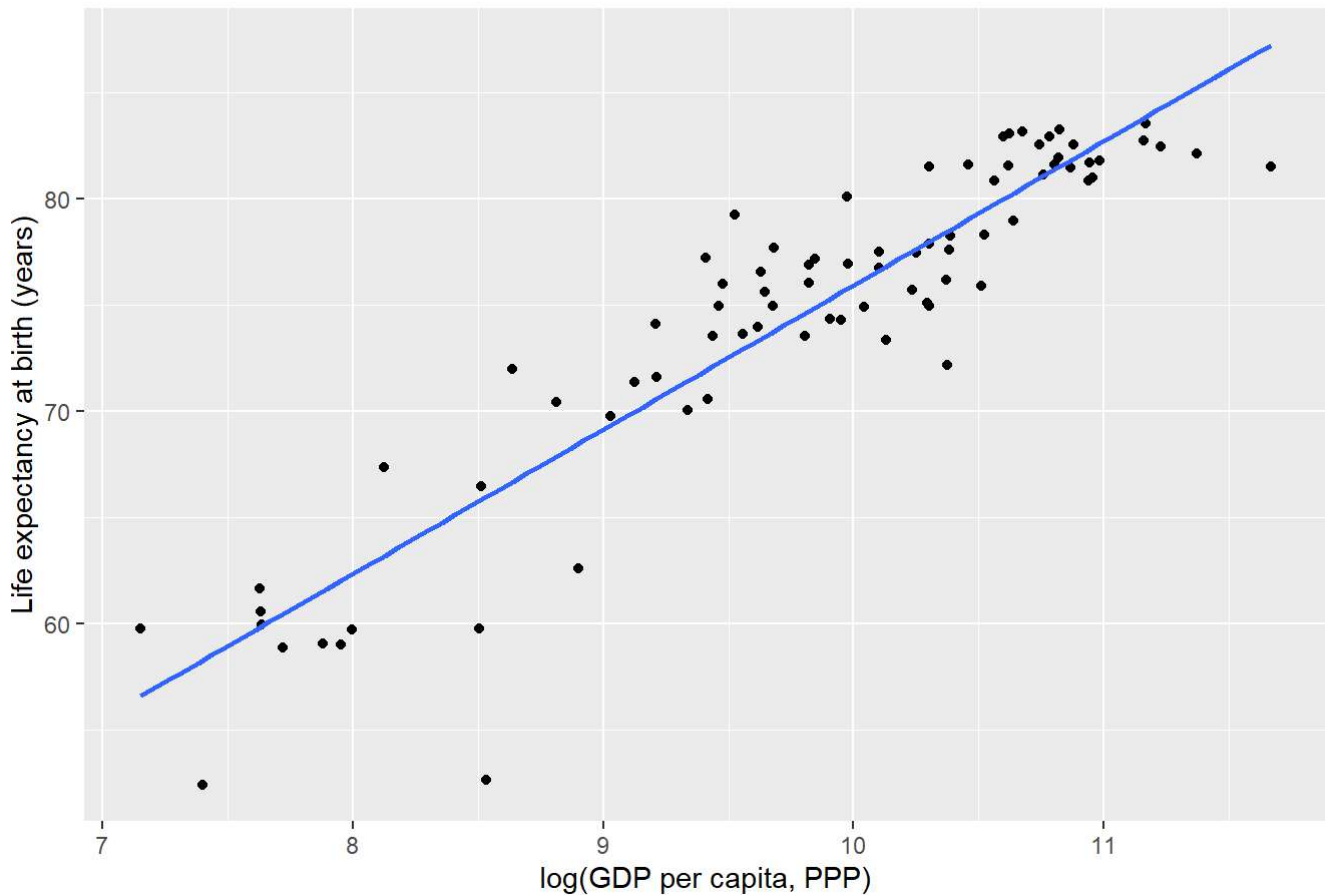
`geom\_smooth()` using formula = 'y ~ x'



```
# Scatter: Life expectancy vs log_GDP, chosen year
ggplot(rq1_year, aes(x = log_GDP, y = LifeExp)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "log(GDP per capita, PPP)",
    y = "Life expectancy at birth (years)",
    title = paste0("Life expectancy vs log GDP per capita in ", unique(rq1_year$Year))
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```

Life expectancy vs log GDP per capita in 2018



Model Interpretations

Model 1 (All country-years, year fixed effects)

- Gini coefficient: small, positive, statistically significant.
- Effect size: a 10-point increase in Gini $\rightarrow \approx +0.8$ years life expectancy.
- GDP coefficient: large and strongly positive.
- Model fit: $R^2 \approx 0.77$.

Over 2000–2023, after adjusting for GDP per capita and global time trends, countries with higher inequality have slightly higher life expectancy, although the effect is very small compared with the strong positive association between GDP and life expectancy.

Model 2 (Single Year, 2018)

Gini coefficient: small, positive, **not statistically significant**.

Effect size: a 10-point increase in Gini $\rightarrow \approx +0.9$ years, but with large uncertainty.

GDP coefficient: large and significant.

Model fit: $R^2 \approx 0.85$.

In 2018 alone, the estimated relationship between Gini and life expectancy is weak and statistically uncertain once GDP is controlled for. GDP per capita remains the main predictor of life expectancy levels across countries.

Comparison across models

The estimated effect of income inequality on life expectancy is very similar in magnitude across the two models (approximately 0.08–0.09 years per Gini point), but only statistically significant in the full panel. This difference arises entirely from sample size: the panel includes more than 1,700 country–year observations, providing enough precision to detect a small effect, whereas the single-year cross-section includes only 81 countries and produces a wide confidence interval.

In both models, the substantive conclusion is the same: inequality has at most a small association with life expectancy once GDP per capita is accounted for. In contrast, the association between GDP per capita and life expectancy is large, strongly positive, and highly robust across both specifications.

Overall, the results suggest that economic development, rather than income inequality, explains the majority of cross-country differences in life expectancy.

Statistical Interpretation:

We estimated two ordinary least squares regression models to assess the association between the Gini index of income inequality and life expectancy at birth, controlling for GDP per capita. The primary specification used the complete country–year panel (2000–2023; $N \approx 1,700$), adjusting for global time trends with year fixed effects:

$$\text{LifeExpct} = \beta_0 + \beta_1 \text{Ginict} + \beta_2 \log(\text{GDPct}) + \gamma_t + \epsilon_{ct}.$$

In this model, the estimated coefficient on the Gini index was positive and statistically significant ($\beta_1 \approx 0.08$, $p < 0.001$). Substantively, a 10-point increase in the Gini index was associated with approximately 0.8 additional years of life expectancy, conditional on GDP per capita and year. The coefficient on log GDP per capita was large and highly significant ($\beta_2 \approx 6.53$, $p < 0.001$), indicating that economic development is a major determinant of life expectancy. The model explained 77% of the variation in life expectancy (adjusted $R^2 = 0.77$).

As a robustness check, we re-estimated the model using only the most recent year with maximum data coverage (2018; $N = 81$):

$$\text{LifeExpct} = \alpha_0 + \alpha_1 \text{Ginic} + \alpha_2 \log(\text{GDPc}) + u_c.$$

In this cross-sectional model, the estimated coefficient on the Gini index remained positive but was not statistically distinguishable from zero ($\alpha_1 \approx 0.086$, $p = 0.12$). The coefficient on log GDP per capita was again large and highly significant ($\alpha_2 \approx 6.98$, $p < 0.001$). Model fit remained high (adjusted $R^2 = 0.85$), with GDP per capita accounting for most cross-country variance.

Across both models, the estimated association between inequality and life expectancy was consistently small in magnitude and dwarfed by the much larger effect of GDP per capita. The sign of the inequality coefficient was positive in both specifications.

Conclusion

Using comprehensive UN data on life expectancy, income inequality, and GDP per capita, we find **no evidence that higher income inequality is associated with lower life expectancy once economic development is taken into account**. Across nearly two decades of data, the estimated association between Gini and life expectancy is small, positive, and statistically detectable only in the full panel due to greater precision. In contrast, GDP per capita is a strong and robust predictor of life expectancy across all specifications.

Taken together, these results indicate that **economic development, not income inequality, is the primary factor explaining cross-country differences in life expectancy in recent decades**. Inequality appears to play at most a modest secondary role once GDP per capita is controlled for.

```
# Write all cleaned and analysis-ready datasets to CSV

# 1. Cleaned Gini file (already provided by teammate)
write.csv(gini,
          file = "data/gini_clean.csv",
          row.names = FALSE)

# 2. Cleaned GDP file
write.csv(gdp_clean,
          file = "data/gdp_clean.csv",
          row.names = FALSE)

# 3. Cleaned Life Expectancy file
write.csv(life_clean,
          file = "data/lifeexp_clean.csv",
          row.names = FALSE)

# 4. Full merged country-year panel
write.csv(rq1_panel,
          file = "data/rq1_panel.csv",
          row.names = FALSE)

# 5. Analysis dataset for single-year cross-section
write.csv(rq1_year,
          file = "data/rq1_year.csv",
          row.names = FALSE)
```